IRS-aided Cell-free Dual-Function Radar and Communication System

Zheng-Ming Jiang, Min Huang, Qijun Deng
The College of Information Engineering, Shenzhen University, Shenzhen, China
e-mail: jiangzhengming216@163.com, pzhang@szu.edu.cn

Abstract

methods.

Index Terms

Frequency diverse array, angle-range-dependent beampattern, random frequency offset, mainlobe interference suppression

I. Introduction

PHASE-array channel

$$\mathbf{H}[k] = \sqrt{\frac{N_t N_r}{P}} \sum_{p=1}^{P} \alpha_p e^{-j2\pi k f_{d,p} \tau_p} \mathbf{a}_r(\theta_{r,p}) \mathbf{a}_t^H(\theta_{t,p}) \tag{1}$$

$$= \mathbf{A}_r \mathbf{A}[k] \mathbf{A}_t^H \tag{2}$$

transmit signal

$$\mathbf{s}[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \mathbf{X}[k] e^{j2\pi nk/N}, \quad n = 0, 1, \ldots, N-1 \tag{3}$$

where is frequenct to time

$$\mathbf{x}[n] = \mathcal{Q}(\mathbf{s}[n]) = \mathrm{sgn}(\Re\{\mathbf{s}[n]\}) + j \cdot \mathrm{sgn}(\Im\{\mathbf{s}[n]\}) = \alpha \mathbf{s}[n] + d_l[n] \tag{4}$$

Performing FFT on the decomposition yields the frequency-domain relationship: we can obtain that

$$\mathbf{X}_{\text{quant}}[l, k] = \alpha \mathbf{X}[l, k] + \mathbf{D}[l, k] \tag{5}$$

receiver siganl (time) At the receiver, the received signal vector at the l l-th symbol and the n n-th sampling instant is:

$$\mathbf{y}_l[n] = \sum_{p=1}^{P} \beta_p \, e^{j2\pi f_{\mathrm{d}p}(T_{\mathrm{sym}} + nT_s/N)} \, \mathbf{a}_r(\phi_{r,p}) \, \mathbf{a}_t^{\mathsf{H}}(\phi_{t,p}) \, \mathbf{x}_l \left[n - \lfloor \tau_p / T_s \rfloor \right] + \mathbf{w}_l[n] \tag{6}$$

where

- $T_{sym} = T_s + T_{cp}$ is time of total symbol
- $T_s = 1/(N\Delta f)$ is the sampling interval,
- $\mathbf{a}_t(\phi_{t,p}) \in \mathbb{C}^{N_t \times 1}$ is the transmit array steering vector,
- $\mathbf{a}_r(\phi_{r,p}) \in \mathbb{C}^{N_r \times 1}$ is the receive array steering vector,
- $\mathbf{w}_l[n] \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ is additive white Gaussian noise.

After removing the cyclic prefix for each receive antenna and each symbol, perform FFT:

$$\mathbf{Y}[l, k] = \sum_{n=0}^{N-1} \mathbf{y}_l[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \ldots, N-1 \tag{7}$$

This work was supported by the Natural Science Foundation of China under Grants U1501253, 61401284, 61471365, 61501485, 61601300, the Natural Science Foundation of Guangdong Province under Grant 2015A030311030, the Foundation of Shenzhen under Grants ZDSYS201507081625213, KC2015ZDYF0023A, JCYJ20140418091413566, 827-000071, JCYJ20150324140036835, the China Postdoctoral Science Foundation Grant 2015M582413, 2015M582414.

Substitute the expression of and use the Bussgang decomposition of

$$Y[l,k] = \sum_{p=1}^{P} \beta_p \cdot e^{j2\pi f_{d_p} l T_{\text{sym}}} \cdot \mathbf{a}_r(\phi_{r,p}) \mathbf{a}_t^H(\phi_{t,p}) \cdot [\alpha \mathbf{X}[l,k] e^{-j2\pi k \Delta f \tau_p} + \mathbf{D}[l,k] e^{-j2\pi k \Delta f \tau_p}] + \mathbf{W}[l,k] \tag{8}$$

$$= \alpha \sum_{p=1}^{P} \beta_p e^{j2\pi f_{d,p} l T_{\text{sym}}} e^{-j2\pi k \Delta f \tau_p} \mathbf{a}_r(\phi_{r,p}) \mathbf{a}_t^H(\phi_{t,p}) \mathbf{X}[l,k] + \tag{9}$$

$$+ \sum_{p=1}^{P} \beta_p e^{j2\pi f_{d_p} l T_{\text{sym}}} e^{-j2\pi k \Delta f \tau_p} \mathbf{a}_r(\phi_{r,p}) \mathbf{a}_t^H(\phi_{t,p}) \mathbf{D}[l,k] + \mathbf{W}[l,k] \tag{10}$$

Maximum unambiguous range

$$R_{\text{max}} = \frac{c}{2\Delta f} \tag{11}$$

Range resolution:

$$\Delta R = \frac{c}{2N\Delta f} \tag{12}$$

Maximum unambiguous velocity:

$$v_{\text{max}} = \frac{\lambda}{4T_{\text{sym}}} \tag{13}$$

$$\Delta v = \frac{\lambda}{2LT_{\text{sym}}} \tag{14}$$

Velocity resolution:
Substitute the time-domain expression
Key Characteristics: Correlation with the signal:
D[l,k] is correlated with
Identical spatial-range-velocity structure: The distortion term also contains the same steering vectors and phase terms
Causes an error floor: Performance is limited even at high SNR p is path

$$y_l(t) = \alpha \cdot s_l(t - \tau) \cdot e^{j2\pi f_d t} \tag{15}$$

where

$$s_l(t) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_{l,k} \cdot e^{j2\pi k \Delta f t}, \quad 0 \le t \le T_s \tag{16}$$

L is the , K is subcarray
radar:

II. Frequency Offset And Beampattern For FDA

III. Conclusion

References

[1] B. Liao, K. Tsui, S. Chan, Robust beamforming with magnitude response constraints using iterative second-order cone programming, *IEEE Trans. Antennas Propag.*, vol. 59, no. 9, pp. 3477C3482, Sep. 2011.